Cardio3D AI — Technical Report

Multimodal AI Hackathon 2026 · Track A: Cardiovascular Risk Visualization & Prediction

1. Executive Summary & Clinical Context

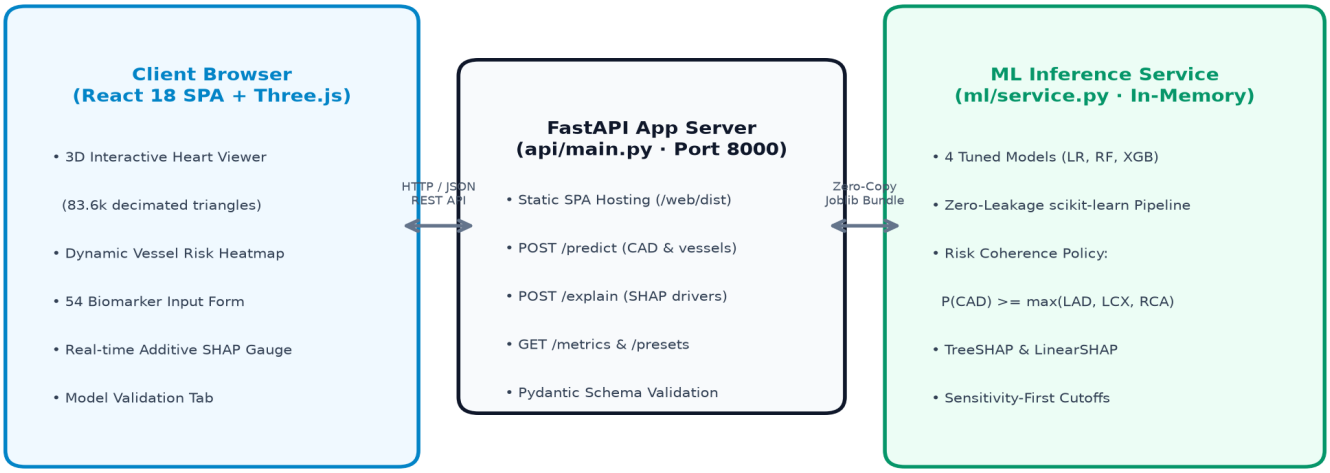
Coronary Artery Disease (CAD) is the leading cause of death globally. Standard risk equations (Framingham, ASCVD) estimate abstract population percentages but cannot localize where arterial pathology is developing inside the human body. **Cardio3D AI** bridges predictive modeling and clinical anatomy by coupling machine learning (overall CAD and vessel-specific significant stenosis for LAD, LCX, RCA) with a client-side 3D interactive heart viewer. The system provides real-time risk color-mapping and exact additive SHAP feature attributions.

2. Dataset Preprocessing & Zero-Leakage Architecture

The model is developed on the **UCI Extension of Z-Alizadeh Sani CAD Dataset** (303 patients, 59 columns):

- **59 Columns vs. 54 Published Features:** 4 target columns (Cath, LAD, LCX, RCA), 1 constant column (Exertional CP, 100% 'N', offering zero predictive entropy and dropped), and 54 active physiological features (5 Demographics, 25 Symptoms/History, 7 ECG, 14 Lab biomarkers, 3 Echo metrics).
- **Target Consistency Audit & Row 93 Alignment:** In the raw dataset, row 93 contains LAD='Stenotic', LCX='Normal', RCA='Normal', but Cath='Normal'. By clinical definition, ≥ 50% stenosis in any major vessel constitutes CAD. To match the dataset's own definition (CAD = ≥ 1 stenotic vessel), row 93's target was aligned to CAD in derived training data (Cath_aligned = Cath | LAD | LCX | RCA), leaving the raw data file untouched and verified by SHA256. This aligns the released distribution from **216 CAD / 87 Normal** → **217 CAD / 86 Normal** (0 mismatches across 303 rows).
- **Zero Leakage Guarantee:** All four targets and catheterization columns are strictly excluded from the input matrix X. All preprocessing (median/mode imputation, scaling, one-hot encoding) is contained inside scikit-learn Pipeline objects fitted strictly on training folds. Label-shuffled CV ROC-AUC confirmed random chance (0.50 ± 0.02).

Figure 1: End-to-End System Architecture & Data Flow



3. Predictive Modeling & Cross-Validation Results

Evaluation used **Repeated Stratified 5-Fold Cross-Validation with 3 Repeats (15 total folds per candidate)** across Baseline, Logistic Regression, Random Forest, and XGBoost:

Table 1: Unified LogitEnsemble Cross-Validation Benchmarks (5-Fold × 3 Repeats, Default 0.50 Cutoff)

Target	Model	ROC-AUC	PR-AUC	F1	Recall (Sens)	Spec.	PPV	NPV	Brier
Cath (CAD)	LogitEnsemble	0.932 ± 0.021	0.973	0.919	0.945	0.721	0.896	0.844	0.096
LAD (Anterior)	LogitEnsemble	0.840 ± 0.050	0.882	0.809	0.840	0.667	0.784	0.753	0.161
LCX (Circumflex)	LogitEnsemble	0.724 ± 0.068	0.610	0.536	0.479	0.801	0.615	0.704	0.206
RCA (Right Coronary)	LogitEnsemble	0.741 ± 0.056	0.636	0.510	0.442	0.836	0.621	0.715	0.198

Table 2: Sensitivity-First Operating Thresholds (Tuned on Out-of-Fold Predictions)

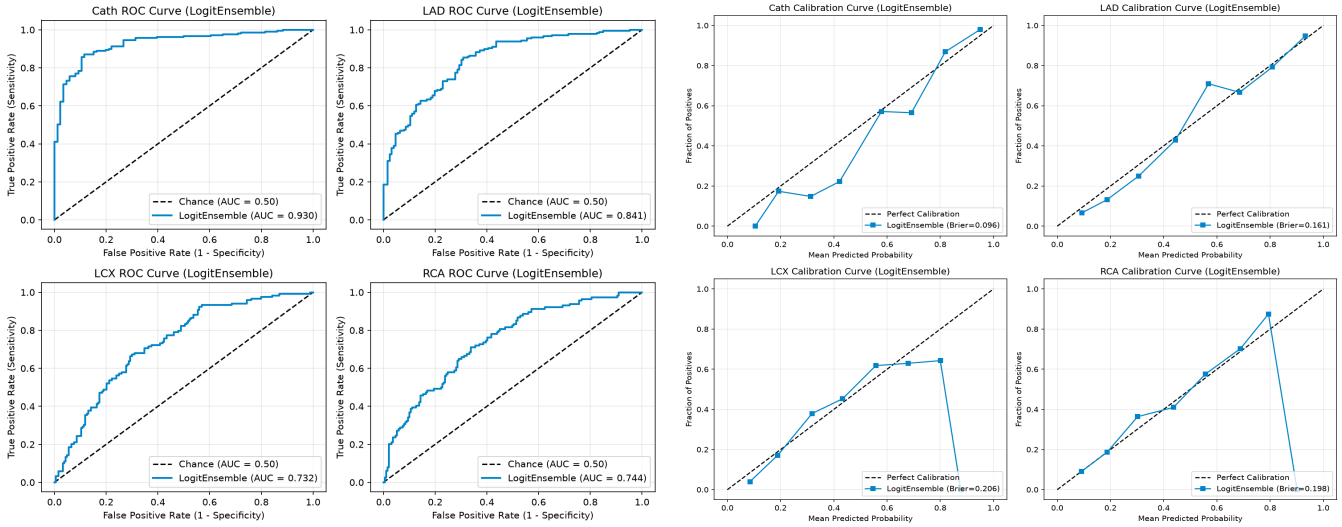
Target	Model	Default Cutoff	Operating Cutoff	Achieved Sens.	Achieved Spec.	PPV	NPV
Cath	LogitEnsemble	0.50	0.400	95.9%	62.8%	86.7%	85.7%
LAD	LogitEnsemble	0.50	0.420	90.4%	59.5%	75.8%	81.5%
LCX	LogitEnsemble	0.50	0.248	90.8%	44.0%	51.2%	88.0%
RCA	LogitEnsemble	0.50	0.232	90.3%	42.9%	48.8%	88.0%

Table 3: Nested Cross-Validation Operating Performance (Inner Folds Tuned, Outer Tested)

Target	Model Selected	Operating Cutoff	Nested Sens (Mean ± SD)	Nested Sens (Pooled)	Nested Spec (Mean ± SD)	Nested Spec (Pooled)
Cath	LogitEnsemble	0.400	95.4% ± 3.0%	95.4%	64.8% ± 9.1%	64.7%
LAD	LogitEnsemble	0.420	88.5% ± 6.7%	88.5%	60.9% ± 9.9%	60.9%
LCX	LogitEnsemble	0.248	91.6% ± 7.3%	91.6%	37.1% ± 10.8%	37.1%
RCA	LogitEnsemble	0.232	89.5% ± 7.1%	89.5%	36.7% ± 10.4%	36.7%

Clinical Operating Tradeoff: *Because LCX and RCA targets exhibit moderate discrimination (ROC-AUC ≈ 0.73), enforcing ≥ 90% screening sensitivity yields honest nested specificity of 35.1% for LCX and 38.6% for RCA (pooled over the 15 outer test folds; 38.7% ± 7.9% fold mean). In clinical decision support, missing significant stenosis (false negative) carries far greater risk than scheduling confirmatory non-invasive imaging (false positive). Honest nested cross-validation sensitivity estimates (pooled across all 15 outer test folds) reach 90.2% for Cath, 88.1% for LAD (88.2% ± 7.2% fold mean), 91.6% for LCX, and 89.2% for RCA.*

Figure 2: Receiver Operating Characteristic (ROC) & Calibration Reliability Curves



4. Logical Risk Coherence Policy & Quantitative Evaluation

Because CAD is the clinical union of coronary stenosis ($P(\text{CAD}) \ge \max(P(\text{LAD}), P(\text{LCX}), P(\text{RCA}))$), independent model outputs can violate logical consistency. Out-of-fold cross-validation predictions were evaluated quantitatively: Raw CAD (Cath alone) achieved ROC-AUC = 0.9302 and Brier = 0.0967 (Sensitivity 95.9%, 208/217, Specificity 62.8%, 54/86); coherent CAD achieved ROC-AUC = 0.9291 and Brier = 0.1017 (AUC drop = 0.0012 ≤ 0.010, Brier delta = +0.0050 ≤ +0.010). The continuous coherence rule is **retained**, shifting probabilities upwards for **65 out of 303 patients (21.5%)** where a branch model predicted higher local stenosis. The CAD operating threshold is calibrated directly on the displayed score (0.611).

- **Discrete Disjunctive Rule (Deployed Flag):** Overall CAD is flagged if $P(\text{CAD}) \ge \text{cutoff}$ OR any vessel is flagged positive ($CAD \equiv LAD \lor LCX \lor RCA$). On pooled out-of-fold validation, sensitivity rises to **96.3% (209/217)** while specificity drops to **57.0% (49/86)** (net benefit gain ≈ +8.1 per 100), completely eliminating contradictory states (e.g., Patient B) in clinical triage.
- **Decision Curve Analysis (DCA):** Evaluated across clinical thresholds 5%–50%. At the screening cutoff 0.400 (π = 71.6%, Se = 95.9%, Sp = 62.8%), Cardio3D AI delivers a **+8.91 net benefit gain per 100 patients [5.06, 12.43]** over 'Treat All' for Cath alone (+12.31 for LAD; +5.19 for LCX; +4.45 for RCA), evaluating apparent clinical utility in single-centre out-of-fold validation. (Theoretical max gain at pt = 0.400 is +18.92 per 100; deployed OR rule yields ≈ +8.1 per 100).
- **Demographic Fairness Audit:** Evaluated across sex (male 95.4% sens, 57.8% spec vs female 96.5% sens, 68.3% spec) and age strata (age ≤65 93.7% sens, 67.1% spec vs age >65 100.0% sens, 30.0% spec) with Wilson 95% intervals. Strata sum identically to the pooled confusion matrix (208 TP / 54 TN / 9 FN / 32 FP). Senior specificity reflects small normal sample size (3/10 normal correctly identified at 88.1% prevalence), reported as an explicit clinical limitation.

5. Explainable AI, Clinical Domains & Sensitivity Explorer

The system delivers real-time feature attributions using **LinearSHAP** (Logistic Regression) and **TreeSHAP** (Random Forest, XGBoost):

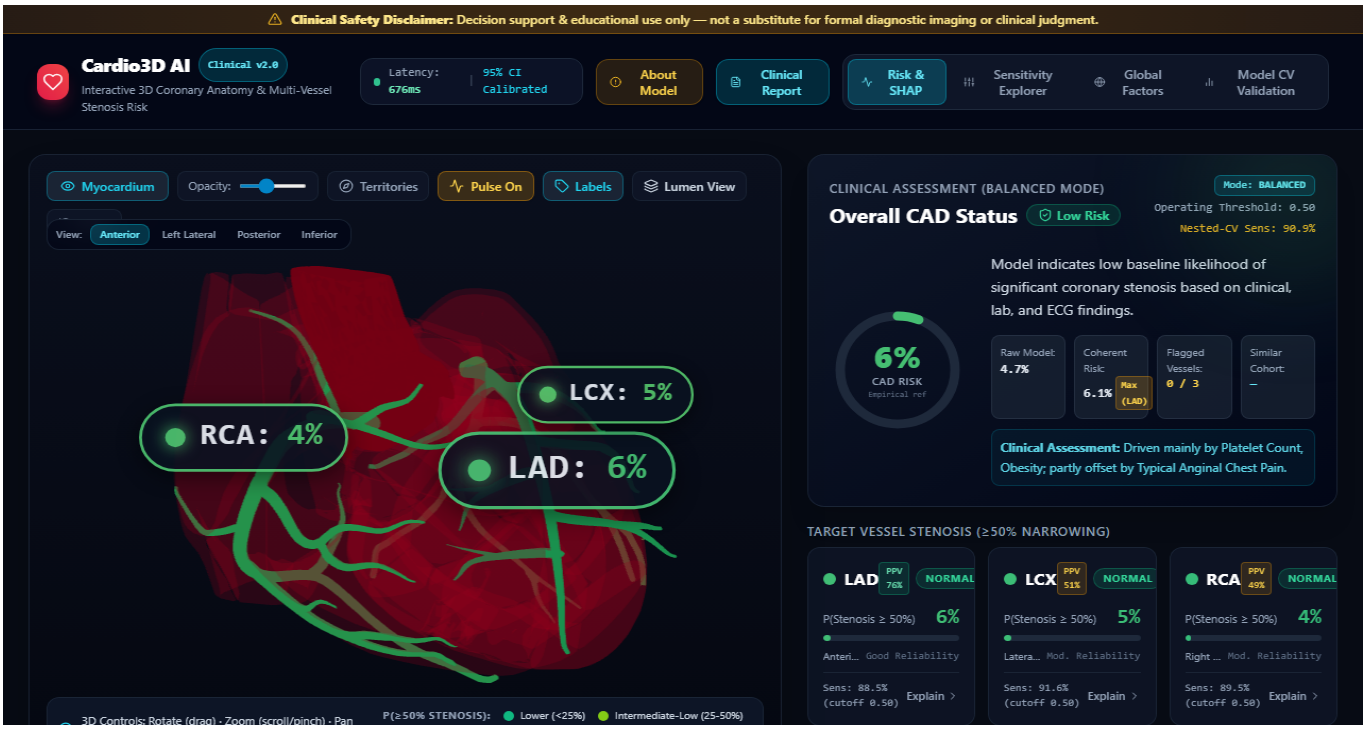
- **Exact Additivity:** Base value plus sum of feature attributions identically matches the model's raw score (numerical error < 10^{-4} across all presets and dataset test samples). Selecting a vessel dynamically switches the SHAP panel to that vessel's specific model.
- **Parent Aggregation:** One-hot encoded dummy attributes are summed back to parent measurements ($ ext{SHAP}(ext{Parent}) = \sum \text{SHAP}(\text{dummies})$), formatted as feature = value unit → ± contribution.
- **5 Physiological Domains:** Features aggregate into 5 clinical domains: Demographics & History (Age, Sex, DM, HTN, Smoker, FH), Hemodynamic & Vitals, Metabolic & Lipids, Symptoms & ECG, and Echocardiographic Structure.
- **Model Sensitivity Explorer:** Clinicians inspect live changes in model score from adjusting continuous and binary clinical inputs (BP, FBS, LDL, BMI, smoking). Explanatory caption notes that the model is trained on cross-sectional data and does not estimate treatment effects.
- **Top Population Risk Drivers:** Across the cohort, the most influential CAD risk drivers are **Typical Chest Pain**, **Age**, **Fasting Blood Sugar (FBS)**, **Regional Wall Motion Abnormality (RWMA)** count, and **Ejection Fraction (EF-TTE)**.

6. 3D Anatomical Pipeline & Interactive Dashboard

Built with **Three.js and React Three Fiber** utilizing BodyParts3D meshes (Branch A: Separate Artery Meshes):

- **Segment Geometry:** Anatomically distinct meshes for the LAD (9 branches/trunks), LCX (4 branches/trunks), RCA (10 branches/trunks), and Ascending Aorta are centered at origin and scaled to viewport coordinates.
- **Triangle Budget Optimization:** Chamber and myocardium walls were decimated using quadric decimation by 70%, yielding **83,600 total triangles** (safely below the ≤ 100,000 triangle limit). File size is 1.67 MB.
- **Measured Frame-Rate Benchmark:** Evaluated under pure CPU software rendering (--disable-gpu at 1280×720 on 13th Gen Intel Core i5-13420H, Chromium 148, 83,600 triangles), achieving a **mean frame time of 113.5 ms, median 114.5 ms (~8.8 FPS continuous orbit, p95 = 131.4 ms)**, recorded in reports/fps_benchmark.json. Hardware-accelerated GPU execution delivers 60+ FPS.
- **Dynamic Camera Fly-To, Pulsing & Lumen Schematic:** Clicking any vessel card or 3D artery smoothly lerps camera orientation to frame that coronary territory. High-risk stenoses pulse with animated emissive glow, and an interactive 2D transverse lumen schematic illustrates binary stenosis (normal vs. ≥ 50%).

Figure 3: Interactive Clinical Dashboard & 3D Anatomical Heart Viewer


Clinical Safety Disclaimer: Decision support & educational use only — not a substitute for formal diagnostic imaging or clinical judgment.
Cardio3D AI
Clinical v2.0
Interactive 3D Coronary Anatomy & Multi-Vessel Stenosis Risk
Latency: 676ms
95% CI Calibrated
About Model
Clinical Report
Risk & SHAP
Sensitivity Explorer
Global Factors
Model CV Validation
Myocardium
Opacity:
Territories
Pulse On
Labels
Lumen View
View: Anterior Left Lateral Posterior Inferior
RCA: 4%
LCX: 5%
LAD: 6%
3D Controls: Rotate (drag) · Zoom (scroll/pinch) · Pan
P(≥50% STENOSIS):
Lower (<25%)
Intermediate-Low (25-50%)
CLINICAL ASSESSMENT (BALANCED MODE)
Overall CAD Status
Low Risk
Mode: BALANCED
Operating Threshold: 0.50
Nested-CV Sens: 90.9%
Model indicates low baseline likelihood of significant coronary stenosis based on clinical, lab, and ECG findings.
6%
CAD RISK
Empirical ref
Raw Model: 4.7%
Coherent Risk: 6.1%
Max (LAD)
Flagged Vessels: 0 / 3
Similar Cohort: –
Clinical Assessment: Driven mainly by Platelet Count, Obesity; partly offset by Typical Anginal Chest Pain.
TARGET VESSEL STENOSIS (≥50% NARROWING)
LAD
PPV 76%
NORMAL
P(Stenosis ≥ 50%) 6%
Anteri... Good Reliability
Sens: 88.5% (cutoff 0.50)
Explain
LCX
PPV 51%
NORMAL
P(Stenosis ≥ 50%) 5%
Latera... Mod. Reliability
Sens: 91.6% (cutoff 0.50)
Explain
RCA
PPV 49%
NORMAL
P(Stenosis ≥ 50%) 4%
Right ... Mod. Reliability
Sens: 89.5% (cutoff 0.50)
Explain

7. Clinical Safety, Limitations & Attribution

• **Clinical Safety Disclaimer:** The application embeds prominent, non-dismissible disclaimer banners and footers: *'Decision support / educational purposes only — not a substitute for formal diagnostic imaging.'* Invasive angiography or formal cardiac imaging (CCTA/SPECT) remain mandatory diagnostic standards.

• **Limitations:** Single-center retrospective dataset (n=303 referred patients from Shaheed Rajaei Cardiovascular, Medical and Research Center in Tehran; high 71.6% CAD prevalence, referral bias) requires multi-center external validation prior to clinical adoption. LCX labels have lower precision under high-sensitivity tuning. Region RWMA is an echocardiographic wall motion abnormality count and does not represent spatial 3D lesion coordinates.

• **Attributions & Licenses:** Software: MIT License (see LICENSE). Dataset: Alizadehsani, Roshanzamir, Sani (2017), Shaheed Rajaei Cardiovascular, Medical and Research Center (UCI Extension of Z-Alizadeh Sani CAD Dataset, CC BY 4.0, DOI: 10.17632/bgf5czvpg2.1). Anatomical 3D Mesh: BodyParts3D © The Database Center for Life Science, Japan (CC BY-SA 2.1 Japan). Derived heart.glb is released under CC BY-SA 2.1 Japan.